

Part I short notes

Chapter one : Set , graph and tree

What We Have Learned So Far

1. A string is defined as a sequence of symbols of finite length.
2. A set is a well-defined collection of objects.
3. The operations union, intersection, difference, complement, Cartesian product etc. can be performed on set.
4. A relation R is called as an equivalence relation on 'A' if R is reflexive, symmetric, and transitive.
5. A set is closed (under an operation) if and only if the operation on two elements of the set produces another element of the set.
6. A graph is an abstract representation of a set of objects where some pairs of the objects are connected by directed or undirected links.
7. A tree is an undirected connected graph with no circuit.
8. A binary tree is a tree in which each node has at most two children.
9. A full binary tree is a tree in which every node other than the leaves has two children.
10. The operations of synchronous circuit are controlled by clock pulses. The operations in an asynchronous circuit are controlled by a number of completion and initialization signals. Here, the completion of one operation is the initialization of the execution of the next consecutive operation.
11. The circuit whose output depends on the present state only is called a combinational circuit., i.e., $O/P = \text{Func.}(\text{Present I/P})$.
12. If the output is the function of the external input and the present stored information, then the circuit is called a sequential circuit., i.e., $O/P = \text{Func.}(\text{External I/P and Present stored information})$.

Chapter 2: Finite Automata

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1. An FA has a finite number of states.
 2. An FA has five characteristics: (i) input, (ii) state, (iii) state transition, (iv) output, and (v) output relation.
 3. Finite automata M is represented as $M = \{Q, \Sigma, \delta, q_0, F\}$.
 4. A string is declared accepted by an FA if the string is finished and the machine has reached a final state.
 5. Finite automata can be of two types (i) deterministic finite automata (DFA) and (ii) non-deterministic finite automata (NFA).
 6. For a DFA, for all cases for a single input given to a single state, the machine goes to a single state, i.e., $Q \times \Sigma \rightarrow Q$.
 7. For an NFA, for some cases for a single input given to a single state, the machine goes to more than one states, i.e., $Q \times \Sigma \rightarrow 2^Q$, where 2^Q is the power set of Q .
 8. NFA can be converted to an equivalent DFA.
 9. An FA is called an NFA with null move if there exists a null transaction, i.e., $Q \times \epsilon \rightarrow Q$.
 10. A dead state is a state where the control can enter and be confined till the input ended, but there is no way to come out from that state.
 11. In a Mealy machine, the output depends on the present state and the present input.
 12. In a Moore machine, the output depends only on the present state.
 13. The Myhill–Nerode theorem is used to minimize finite automata.
 14. Two-way finite automata are machines which can traverse (read) an input string in both directions (left and right).
 15. Finite automata are used in designing lexical analysers.

Chapter 3: Regular expression

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1. An RE can be defined as a language or string accepted by an FA.
 2. Any terminal symbols, null string (Λ), or null set (ϕ) are RE.
 3. Union, concatenation, iteration of two REs, or any combination of them are also RE.
 4. The Arden's theorem states that if P and Q are two REs over Σ , and if P does not contain Λ , then the equation $R = Q + RP$ has a unique (one and only one) solution $R = QP^*$.
 5. The Arden's theorem is used to construct an RE from a given FA by the algebraic method.
 6. If any FA contains any ϵ (null) move or transition, then that FA is called NFA with ϵ -moves.
 7. The ϵ -closure of a state is defined as the set of all states S, such that it can reach from that state to all the states in S with input ϵ (i.e., with no input).
 8. The pumping lemma for RE is used to prove that certain sets are not regular.
 9. A set is closed (under an operation) if and only if the operation on two elements of the set produces another element of the set.
 10. Closure is a property which describes when we combine any two elements of the set, the result is also included in the set.
 11. REs are closed under union, complementation, and intersection operation.

Chapter 4 Pushdown Automata

1. Pushdown automata (in short PDA) is the machine format of context-free language.
2. The mechanical diagram of a pushdown automata contains the input tape, reading head, finite control, and a stack.
3. A pushdown automata consists of a 7-tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ where Q, Σ, q_0 , and F have their original meaning, Γ is finite set of stack symbols, and z_0 is the stack bottom symbol.
4. In a PDA, the transitional function δ is in the form $Q \times (\Sigma \cup \{\lambda\}) \times \Gamma \rightarrow (Q, \Gamma)$.
5. Instantaneous description (ID) remembers the information of state and stack content at a given instance of time.
6. There are two ways to declare a string to be accepted by pushdown automata: (a) accepted by the empty Stack (store) and (b) accepted by the final state.
7. A string is declared accepted by a PDA by the empty stack if the string is totally traversed and the stack is empty.
8. A string is declared accepted by a PDA by the final state if the string is totally traversed and the machine reaches its final state.
9. A string is accepted by a PDA by the empty stack if and only if the language is accepted by that PDA by the final state.
10. If a PDA being in a state with a single input and a single stack symbol gives a single move, then the PDA is called deterministic pushdown automata.
11. If a PDA being in a state with a single input and a single stack symbol gives more than one move for any of its transitional functions, then the PDA is called non-deterministic pushdown automata.
12. From context-free grammar, the NPDA can be directly constructed.

Complexity Theory

Turing Machines

1. The Turing machine was proposed by A.M. Turing in 1936 as a model of any possible combination. Any computational process carried out by the present day's computer can be done on the Turing machine.
2. The Turing machine is the machine format of unrestricted language, i.e., all types of languages are accepted by the Turing machine.
3. Undecidable problems are not solved by the computer as no Turing machine can be developed for these problems.
4. The mathematical description of the Turing machine consists of 7 tuples $(Q, \Sigma, \Gamma, \delta, q_0, B, F)$, where Q is the finite set of states, Σ is the finite set of input alphabets, Γ is the finite set of allowable tape symbol, δ is the transitional function, q_0 is the initial state, B is the blank symbol, and F is the final state.
5. The transitional functions of the Turing machine are in the form $Q \times \Gamma \rightarrow (Q \times \Gamma \times \{L, R, H\})$. That is, from one state by getting one input from the input tape the machine moves to a state, writing a symbol on the tape, and moves to left, right, or halts.
6. The mechanical diagram of the Turing machine consists of the input tape, the finite control and the read-write head.
7. Upon execution of a transitional function, the machine goes to some state, writes a symbol in the cell of the input tape from where the input symbol was scanned, and moves the reading head to the left or right or halts.
8. The instantaneous description (ID) of a Turing machine remembers the contents of all cells from the rightmost to at least the leftmost, the cell currently being scanned by the read-write head, and the state of the machine at a given instance of time.

Turing variations

1. A multi-tape Turing machine has multiple tapes and can be converted to a single-tape Turing machine
2. A multi-tape Turing machine is less complex than a single-tape Turing machine.
3. A multi-head Turing machine has multiple heads attached with a single finite control.
4. Every multi-head Turing machine has an equivalent single-head Turing machine.
5. A Turing machine where there are infinite numbers of sequence of blank on both sides of the tape is called a two-way infinite tape Turing machine.
6. The input tape of a multi-dimensional Turing machine is extended to infinity in more than one dimension.
7. An enumerator is a type of Turing machine attached to a printer.
8. The allowable input tape space of the linear bounded automata is the linear function of the length of input string.
9. A Post machine has a queue in comparison with a PDA which has a stack.

Part II Questions and Answers

Fill in the Blanks

1. For a string, any prefix of the string other than the string itself is called as the _____ of the string.
2. For a string, any suffix of the string other than the string itself is called as the _____ of the string.
3. If there are two sets A and B, then their intersection is denoted by _____.
4. For a set of elements n, the number of elements of the power set of A is _____.
5. A relation R is said to be _____, if for two elements 'a' and 'b' in X, if a is related to b then b is related to a.
6. A relation R is said to be _____, if for a, b, c $\in$ A and if aRb, bRc hold good then aRc also holds good.
7. A relation R is called as an _____ on 'A', if R is reflexive, symmetric, and transitive.

Answers:

1. Proper prefix

4. 2^n

7. equivalence relation

2. proper suffix

5. symmetric

3. $A \cap B$

6. transitive

1. Grammar consists of four tuples—Set of non-terminals, _____, set of production rule, and _____.
2. According to the Chomsky hierarchy, there are _____ types of grammars.
3. Type 1 grammar is called _____.
4. Type 2 grammar is called _____.
5. According to the Chomsky hierarchy, regular grammar is type _____ grammar.
6. All languages are accepted by _____.
7. The machine format of context-free language is _____.
8. Linear bounded automata is the machine format of _____.
9. The machine format of type 3 language is _____.
10. Grammar where production rules are in the format $\alpha A \beta \rightarrow \alpha \gamma \beta$ is _____ grammar.
11. In a context-free grammar at the left hand side, there is _____ non-terminal.
12. Type 3 language is called _____.
13. $a^n b^n c^n$ is an example of _____ language in particular.
14. The grammar $S \rightarrow aSb/A, A \rightarrow Ac/c$ is an example of _____ grammar in particular.
15. The grammar $S \rightarrow Abc/ABSc, BA \rightarrow AB, Bb \rightarrow bb, A \rightarrow a$ is an example of _____ grammar in particular.
16. The grammar $A \rightarrow aA/bB/a/b, B \rightarrow bB/b$ is an example of _____ grammar in particular.
17. The language $a^*(a + b)b^*$ is an example of _____ language in particular.
18. $L =$ Alternating sequence of 'a' and 'b' is an example of _____ language in particular.
19. Regular expression is accepted by type _____ grammar.

Answers:

1. set of terminals, start symbol
4. Context-free grammar
7. Push down automata
10. Context-sensitive
13. Context-sensitive
16. Regular grammar
19. Three

2. Four
5. Three
8. Context-sensitive grammar
11. Single
14. Context-free
17. Type 4

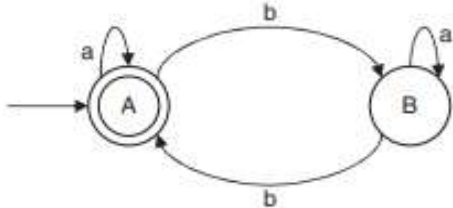
3. Context-sensitive grammar
6. Turing machine
9. Finite automata
12. Regular expression
15. Context-sensitive
18. Context-free

1. Finite automata is defined as $M = \{ \text{---}, \text{---}, \text{---}, \text{---}, \text{---} \}$.
2. Finite automata is one type of _____ state machine.
3. In the graphical format representation of finite automata, by double circle _____ state is represented.
4. In the tabular format representation of finite automata, by single circle _____ state is represented.
5. In the block diagram of finite automata, the input from input tape is read by _____.
6. Finite control can be considered as a _____ unit of a finite automaton.
7. If the input is given null for a state of finite automata, the machine goes to the _____.
8. There are two conditions for declaring a string accepted by a finite automaton. The conditions are _____ and _____.
9. For DFA, the transitional function δ is represented by $Q \times \Sigma \rightarrow$ _____.
10. DFA is a finite automata where for all cases for a single input given to a single state, the machine goes to a _____ state.
11. In NFA, δ is a transitional function mapping $Q \times \Sigma \rightarrow$ _____.
12. If any finite automata contains any ϵ (null) move or transaction, then that finite automata is called _____ with ϵ moves.
13. A state where the control can enter and be confined till the input ended, but there is no way to come out from that state is called _____.
14. The Mealy machine and Moore machine are example of finite automata with _____.
15. For the Mealy machine, the output depends on _____ and _____.
16. For the Moore machine, the output depends on _____.
17. The states which can never be reached from the initial state are called _____ states.
18. A relation R in set S is called equivalence relation if it is _____, _____, and _____.
19. A relation R in set S is _____ if for x, y in S, yRx whenever xRy .
20. A relation R in set S is _____ if for x, y, and z in S, xRz whenever xRy and yRz .
21. R is _____ then for all x, y, and z, $xRy \Rightarrow xzRyz$.
22. The Myhill–Nerode theorem is used for _____.
23. A two-way finite automata is like finite automata but can traverse in _____.
24. The lexical analyser is designed by _____.
25. The maximum number of states of a DFA converted from an NFA with n states is _____.
26. A finite automata recognizes _____ language.

Answers:

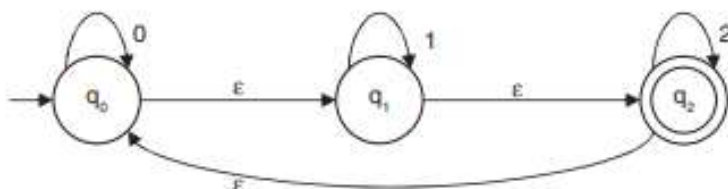
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|--------------------------------|---|---------------------|--------------------------------------|-------------|
| 1. $Q, \Sigma, \delta, q_0, F$ | 2. finite | 11. 2^n | 20. transitive | 26. regular |
| 4. final | 5. reading head | 12. NFA | 21. right invariant | |
| 7. same state | 8. string must be totally traversed, the machine must come to a final state | 13. dead state | 14. outputs | |
| 10. single | | 16. present state | 15. present state, present input | |
| 19. symmetric | | 17. inaccessible | 18. reflexive, symmetric, transitive | |
| 22. minimization of FA | | 20. transitive | 21. right invariant | |
| 23. both directions | | 23. both directions | 24. finite automata | |

Multiple Choice Questions

1. The machine format of regular expression is
 - a) Finite automata
 - b) Push down automata
 - c) Turing machine
 - d) All of the above
2. The regular expression is accepted by
 - a) Finite automata
 - b) Push down automata
 - c) Turing machine
 - d) All of the above
3. The language of all words with at least 2 a's can be described by the regular expression
 - a) $(ab)^*a$
 - b) $(a+b)^*ab^*a(a+b)^*$
 - c) $b^*ab^*a(a+b)^*$
 - d) All of the above
4. The set of all strings of $\{0, 1\}$ having exactly two 0's is
 - a) $1^*01^*01^*$
 - b) $\{0+1\}^*1$
 - c) $\{11+0\}^*$
 - d) $\{00+11\}^*$
5. The set of all strings of $\{0, 1\}$ where 0 is followed by 1 and 1 is followed by 0 is
 - a) $(01)^*$
 - b) 0^*1^*
 - c) 0^*1+1^*0
 - d) $0^*1^*+1^*0^*$
6. Which of the strings do not belong to the regular expression $(ba + baa)^*aaba$
 - a) baaaba
 - b) babaabaaaba
 - c) babababa
 - d) baaaaba
7. Which of the following type of language is regular expression?
 - a) Type 0
 - b) Type 1
 - c) Type 2
 - d) Type 3
8. Which of the following RE over $\{0,1\}$ denotes the set of all strings not containing 100 as substring?
 - a) $(1+0)^*0^*$
 - b) 0^*1010^*
 - c) 0^*1^*01
 - d) All of the above
9. $\wedge + RR^* = ?$
 - a) R
 - b) R^*
 - c) R^+
 - d) $\wedge$
10. $\{a/b\}$ denotes the set
 - a) $\{ab\}$
 - b) $\{a, b\}$
 - c) $\{a, b\}^*$
 - d) $\{ab\}^*$
11. What is the solution for the equation $R = Q + RP$ (If P and Q are RE and P does not contain $\wedge$)?
 - a) $R = QP^*$
 - b) $R = QP$
 - c) $R = PQ^*$
 - d) $R = P^*Q^*$
12. Which of the languages is accepted by the following FA?
 
 - a) $b(a + bba^*)^*a^*$
 - b) $a(a + bba^*)^*b^*$
 - c) $a(a + bb^*a^*)^*b^*$
 - d) None of these
13. Which is true for ϵ -closure of a state?
 - a) All the states reachable from the state with input null excluding the state
 - b) The state only
 - c) All the other states
 - d) All the states reachable from the state with input null
14. The pumping lemma for regular expression is used to prove that
 - a) Certain sets are regular
 - b) Certain sets are not regular
 - c) Certain regular grammar produce RE
 - d) Certain regular grammar does not produce RE
15. Regular sets are closed under
 - a) Union
 - b) Concatenation
 - c) Kleene closure
 - d) All of the above

1. a 2. d 3. b 4. a 5. d 6. c 7. c 8. c 9. b 10. b 11. a 12. a 13. d 14. a 15. d

1. Regular expression is a string on _____ among $\{Q, \Sigma, \delta, q_0, F\}$
2. If R is a regular expression, then $\Lambda + RR^* =$ _____
3. According to the Arden's theorem, the solution of the equation $R = Q + RP$ is _____
4. The set of all states that can be reached from that state to all the states with input ϵ is called _____
5. For the following NFA with ϵ -move



ϵ -closure of q_1 is _____.

6. A grammar where all productions are in the form $A \rightarrow B\alpha$ or $A \rightarrow \alpha$ is called _____.
7. A grammar where all productions are in the form $A \rightarrow \alpha B$ or $A \rightarrow \alpha$ is called _____.
8. The pumping lemma for regular expression is used to prove that certain sets are not _____.
9. The property which describes when we combine any two elements of the set and the result is also included in the set is called _____.
10. If any finite automata contains any ϵ (null) move or transaction, then that finite automata is called _____.
11. The machine format of regular expression is _____.

Answers:

1. Σ
2. R^*
3. $R = QP^*$
4. ϵ -closure
5. q_0, q_1, q_2
6. left linear grammar
7. right linear grammar
8. regular
9. closure property
10. NFA with Σ moves
11. Finite automata

1. Context-free language is _____ language.
 - a) Type 0 b) Type 1
 - c) Type 2 d) Type 3
2. Parsing a string from a given grammar means
 - a) Finding a derivation
 - b) Finding a leftmost derivation
 - c) Finding a rightmost derivation
 - d) Finding a derivation tree.
3. A grammar is called ambiguous if
 - a) It generates more than one string
 - b) It generates both leftmost and rightmost derivation for a given string
 - c) It generates more than one parse tree for a given string
 - d) It fulfills both (b) and (c)
4. Which is not true for ambiguous grammar?
 - a) Ambiguity creates problem in generating a language from a given grammar
 - b) All ambiguity can be removed.
 - c) Inherent ambiguity cannot be removed
 - d) Some ambiguity can be removed by hand
5. Non-generating symbols are those symbols which
 - a) Do not generate any string of non-terminals
 - b) Do not generate any null string
 - c) Do not generate any string of terminal and non-terminals
 - d) Do not generate any string of terminals
6. Useless symbols in CFG are
 - a) Non-generating symbols and non-reachable symbols
 - b) Null alphabets and null string
 - c) Non-terminal symbols
 - d) All of these
7. Which of the following is a unit production?
 - a) (String of NT) $\rightarrow$ (String of NT)
 - b) (Single NT) $\rightarrow$ (String of NT)
 - c) (Single NT) $\rightarrow$ (Single NT)
 - d) (String of NT) $\rightarrow$ (Single NT)
8. Which is true for the following CFG?

$$S \rightarrow aA/\epsilon$$

$$A \rightarrow bA/a$$
 - a) Null production can be removed
 - b) Null production cannot be removed
 - c) As A does not produce null, null cannot be removed
 - d) Both (b) and (c)
9. Which of the following production is in CNF? (more specific)
 - a) (NT) $\rightarrow$ (String of NT)
 - b) (NT) $\rightarrow$ (String of terminal and non-terminal)
 - c) (NT) $\rightarrow$ (String of terminal)
 - d) (NT) $\rightarrow$ (String of exactly two NT)
10. Which of the following production is in GNF? (more specific)
 - a) (NT) $\rightarrow$ (Single T)(String of NT)
 - b) (NT) $\rightarrow$ (Single NT)(String of T)
 - c) (NT) $\rightarrow$ (String of terminal and non-terminal)
 - d) (NT) $\rightarrow$ (String of NT)
11. Which of the following is common in both CNF and GNF?
 - a) (NT) $\rightarrow$ (Single T)(String of NT)
 - b) (NT) $\rightarrow$ (String of exactly two NT)
 - c) (NT) $\rightarrow$ (String of NT)
 - d) (NT) $\rightarrow$ (Single T)
12. Context-free language is not closed under
 - a) Union b) Concatenation
 - c) Complementation d) Star closure

1. PDA is the machine format of
 - a) Type 0 language
 - b) Type 1 language
 - c) Type 2 language
 - d) Type 3 language
2. Which is not true for mechanical diagram of PDA?
 - a) PDA contains a stack
 - b) The head reads as well as writes
 - c) The head moves from left to right
 - d) The input string is surrounded by an infinite number of blanks in both sides
3. The difference between finite automata and pushdown automata is in _____.
 - a) Reading head
 - b) Input tape
 - c) Finite control
 - d) Stack
4. In the PDA, transitional function δ is in the form
 - a) $Q \times (\Sigma \cup \{\lambda\}) \times \Gamma \rightarrow (Q, \Gamma)$
 - b) $Q \times \Sigma \rightarrow Q$
 - c) $Q \times \Sigma \times \Gamma \rightarrow Q$
 - d) $Q \times \Gamma \rightarrow Q \times \Sigma$

(Where Q is the finite set of states, Σ is the set of input alphabets, and Γ is the stack symbols.)
5. Instantaneous description remembers
 - a) The information of state and input tape content at a given instance of time.
 - b) The information of state and stack content at a given instance of time.
 - c) The information of input tape and stack content at a given instance of time.
 - d) The information of state, input tape, and stack content at a given instance of time.
6. Which of the following is not possible algorithmically?
 - a) RE to CFG
 - b) NFA to DFA
 - c) CFG to PDA
 - d) NPDA to DPDA

Answers:

1. c
2. b
3. d
4. a
5. b
6. d

1. The full form of PDA is _____.
2. _____ is the machine format of context-free language.
3. The mechanical diagram of pushdown automata is similar to finite automata, but the difference is in _____.
4. _____ of PDA describes the configuration of PDA at a given instance.
5. There are two ways to declare a string accepted by a pushdown automata: (a) _____ and (b) _____.
6. For directly constructing a PDA from a given grammar, we first need to convert the grammar into _____ normal form.
7. If a PDA being in a state with a single input and a single stack symbol gives a single move, then the PDA is called _____ pushdown automata.
8. If a PDA being in a state with a single input and a single stack symbol gives more than one move for any of its transitional functions, then the PDA is called _____ pushdown automata.
9. The PDA for the string $L = \{ \text{Set of all palindromes over } a, b \}$ can be designed by _____ pushdown automata and not by _____ pushdown automata.

Answers:

1. pushdown automata
2. Pushdown automata
3. stack
4. Instantaneous description (ID)
5. accepted by the empty stack,
6. Greibach
7. deterministic
8. non-deterministic
9. non-deterministic, deterministic

Complexity Theory

1. All types of languages are accepted by _____.
2. According to the Chomsky hierarchy, type 0 language is called _____.
3. The diagram of the Turing machine is like finite automata, but here the head moves _____.
4. The head of the Turing machine is called _____.
5. The string $a^n b^n c^n$, $n > 0$ is accepted by _____.
6. The Turing machine is called _____.
7. Two-stack PDA is equivalent to _____.
8. The crash condition occurs if the read-write head of a Turing machine is over the _____.
9. A Turing machine is said to be in _____ if it is not able to move further.

Answers:

1. Turing machine
2. unrestricted language
3. in both direction
4. Read-write head
5. Turing machine
6. universal machine
7. Turing machine
8. leftmost cell
9. halt state

1. Which of the following is true (A : Multi-tape TM, B: Multi-head TM, C: Non- deterministic TM, D: K-dimensional TM, E: Single tape TM).
 a) $A, B > E$ b) $C > E$ c) $D > A, B > C > E$ d) $A = B = C = D = E$
2. Which of the following has a read only tape?
 a) Single-tape TM b) Multi-tape TM
 c) Linear bounded automata d) None of these.
3. Which one of the following conversions is impossible ?
 a) NTM $\rightarrow$ DTM b) Multi-tape TM $\rightarrow$ Single-tape TM
 c) Multi-head TM $\rightarrow$ Single-head TM d) NPDA $\rightarrow$ DPDA
4. The movement of the head is limited to a certain region for which of the following machine
 a) K- dimensional TM b) Linear bounded automata
 c) Pushdown automata d) Two-way infinite tape
5. In which of the cases stated is the following statement true?
 'For every non-deterministic machine M_1 , there exists an equivalent deterministic machine M_2 recognizing the same language.'
 a) M_1 is a non-deterministic finite automata
 b) M_1 is a non-deterministic PDA
 c) M_1 is a non-deterministic TM
 d) For no machine M_1 use the above statement true.

Answers:

1. d 2. c 3. d 4. b 5. a